

PID method-selection, mathematics, and publication contract

A fail-closed scientific-object, applicability, evidence, and process contract for Prisoma and its ecosystem consumers.

There is no generic PID result. A shared label, author, or software route does not make two functionals, quantities, or estimands identical.

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METHOD AND PUBLICATION STAGES

M0 Preserve	M1 Type	M2 Prove	M3 Admit	M4 Design	M5 Freeze	M6 Execute	M7 Challenge	M8 Publish
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Preserve valid novel work. Block inapplicable claims. Never use one PID as an automatic fallback for another.

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This contract governs PID method selection, result identity, interpretation, and publication in Prisoma. It also governs PID-derived work shared with ecosystem projects such as Galadriel. The canonical thesis claims and four scientific gates remain in `grandplan.md`. This document makes the method mathematics and route boundaries explicit.

1. Decision

There is no generic “PID result” and no generic “Wibral PID.” A PID is measure-relative. Its numerical value is not identified until the source and target variables, probability law, functional, output quantity or lattice coordinate, source order, aggregation, component, and units are fixed. A sample result also needs an estimator and sampling design. A declared-law result needs an evaluator instead.

Prisoma must keep these nine kinds of object separate:

- 1 a paper-defined **functional**, including an exact instance of a parameterized functional family, on a declared class of probability laws;
- 2 a **quantity or coordinate** defined within that functional, such as a cumulative lattice value or one Möbius-inverted atom;
- 3 a **declared-law evaluator** that maps one specified law to that functional's named quantity;
- 4 a **sample estimator** that maps observations to an estimate of that named quantity;
- 5 a fitted or fixed **transform** that creates the variables supplied to a route;
- 6 an exact or interval **certifier** with a stated admitted domain;
- 7 a theorem, fixture, or other **validation artifact**;
- 8 an **objective composition** that combines named PID coordinates with coefficients or terms;
- 9 an application-level **interpretation or decision rule**.

One object can cite another. It cannot impersonate it. No scientific conclusion transfers between two functionals, law classes, estimators, transforms, or source counts without an explicit mapping theorem whose assumptions hold in the application.

Valid novel work must be preserved. A route that is not ready for an active Prisoma hypothesis stays in the research registry with its exact definition, status, evidence, and blockers. It is not silently deleted, relabeled as another PID, or used as a fallback.

2. Scientific-object registry

The IDs below are Prisoma semantic IDs. They are not names assigned by the cited authors. Existing `pid-rs` method IDs remain canonical implementation-route IDs. They must not be demoted to aliases for the new functional nodes.

Object	Mathematical domain and role	Current Prisoma role	Prohibited equivalence
<code>functional.shared-exclusions.mgw-categorical</code>	Makkeh–Gutknecht–Wibral pointwise shared exclusions on categorical probability laws; averaging and Möbius inversion give measure-specific atoms	Active low-dimensional categorical functional	Continuous Ehrlich shared exclusions, <code>I_min</code> , BROJA, or an objective
<code>route.shared-exclusions.mgw-empirical-pmf</code>	Plug-in route from sampled categorical rows or positive empirical counts to the MGW functional on the induced empirical law	Active at the pinned row API; sparse count API requested upstream	A specified population law, a certificate of sampling validity, or a second functional

Object	Mathematical domain and role	Current Prisoma role	Prohibited equivalence
functional.shared-exclusions.measure-theoretic-2021	Schick-Poland et al. construction for broad measure-theoretic random-variable domains	Theorem and domain boundary	The practical Ehrlich formula or its kNN estimator
functional.shared-exclusions.ehrlich-continuous	Ehrlich et al. purely continuous analytic shared-exclusions formulation with its own law and gauge assumptions	Default-off research functional	Binned MGW or a proved categorical-to-continuous limit
route.shared-exclusions.ehrlich-source-disjunction-knn	Finite-sample nearest-neighbor route targeting the Ehrlich functional	Default-off and application-blocked	The functional itself, categorical MGW, or generic KSG validity
functional.pid.williams-beer-imin	Williams–Beer categorical redundancy functional	Preserved inactive comparator	Shared exclusions or a rescue after another route abstains
functional.pid.broja-bivariate	Bertschinger et al. optimization-based bivariate PID, also called the $\sim$ -PID; union information and unique-information formulations identify the same decomposition	Preserved inactive comparator	Shared exclusions, $I_{\min}$, deficiency PID, I-PID, or a multi-source PID
functional.pid.sim-g-gaussian-restricted	Venkatesh et al. $\sim$ -G-PID: the BROJA/ $\sim$ optimization with the coupling Q_{MXY} restricted to jointly Gaussian laws	Preserved research functional	BROJA/ $\sim$ itself: $\sim$ -G only bounds it in general, and equality for Gaussian input P is conjectured
evaluator.pid.sim-g-gaussian-covariance-law	Covariance-law optimization that evaluates the $\sim$ -G-PID on an admitted jointly Gaussian law	Future declared-law research route	An evaluator for unrestricted BROJA/ $\sim$, a sample estimate, or proof of the equality conjecture
route.pid.sim-g-gaussian-sample-plug-in	Sample-covariance plug-in estimator targeting $\sim$ -G	Preserved research candidate for justified continuous Gaussian regimes	The declared-law evaluator, a Gaussianity test, BROJA/ $\sim$, or the bias-corrected route
route.pid.sim-g-gaussian-sample-bias-corrected-2023	Distinct finite-sample bias-corrected estimator targeting $\sim$ -G	Preserved research candidate	A new functional, an exact evaluator, unrestricted BROJA/ $\sim$, or permission to analyze atomic spike counts
functional.pid.deficiency-delta-bivariate	Banerjee et al. deficiency-based bivariate δ -PID	Preserved research comparator	BROJA/ $\sim$, $\sim$ -G, MMI, shared exclusions, or I-PID
functional.pid.deficiency-delta-g-gaussian-channel-restricted	Gaussian deficiency δ -G, which restricts the degrading channel in the deficiency optimization to linear additive Gaussian channels	Preserved research functional	The unrestricted δ -PID; the restriction gives an upper bound on deficiency
functional.pid.deficiency-delta-g-hat-convex	The paper-defined δ -G-PID obtained from a further convex surrogate; its atoms bound the unrestricted δ -PID and agree in proved extremal cases	Preserved research proxy functional	δ , δ -G, or mere numerical solver error
evaluator.pid.deficiency-delta-g-hat-convex-law	Declared-Gaussian-covariance route for evaluating the δ -G convex program	Preserved research evaluator route	An exact evaluator of δ or δ -G, a sample estimator, or a certificate
functional.pid.deficiency-lagrangian-delta-lambda-bivariate	Venkatesh–Gurushankar–Schamberg parameterized δ^λ -PID family for $\lambda \geq 0$; the paper states deficiency and BROJA endpoint connections, but the small- λ raw-value limit needs an explicit normalization or lexicographic theorem	Preserved research functional family; every $\lambda > 0$ identifies one instance and $\lambda = 0$ is a degenerate endpoint	One parameter-free functional, an unqualified <code>recovers_on_domain</code> edge, or permission to conflate parameter instances
functional.pid.information-deficiency-i-bivariate	Venkatesh–Gurushankar–Schamberg I-PID based on information deficiency, designed to capture unique information as an auxiliary random variable	Preserved research candidate	BROJA/ $\sim$, δ , δ^λ , or a Gaussian estimator
functional.information-hierarchy.lyu-conditional-independence	Conditional-independence quantities: two-source redundancy, per-source unique information, signed order-K synergistic effects, narrow synergy, and total synergistic effect; for $N \geq 3$ it deliberately defines no redundancy or complete PID lattice	Preserved research candidate	A complete multi-source antichain PID, a nonnegative atom system, or a shared-exclusions decomposition
evaluator.information-hierarchy.lyu-gaussian-covariance-law	Log-determinant evaluation of the named hierarchy quantities on positive-definite jointly Gaussian laws	Future declared-law sensitivity route	A sample estimate or a distribution-free formula
route.information-hierarchy.lyu-sample-covariance-plug-in	Sample-covariance plug-in estimator for those named Gaussian hierarchy quantities	Preserved research candidate	The population evaluator, a full $N \geq 3$ PID, or a validation of another measure
composition.informorphic.bivariate-2025	Paper-defined parametric family of coefficient-weighted compositions of named categorical PID atoms and residual entropy	Future typed composition; each coefficient vector is a distinct objective instance	A PID functional, estimator, declared-law evaluator, or gradient theorem

Object	Mathematical domain and role	Current Prisoma role	Prohibited equivalence
<code>composition.infomorphich.trivariate-2025</code>	Distinct paper-defined parametric family of trivariate local objective compositions	Future typed composition; each atom map and coefficient vector is bound	The bivariate family, a new PID definition, or a coefficient-free objective
<code>diagnostic.shannon-invariant.*</code>	Measure-independent algebraic summary of mutual-information terms	Screening only after every constituent MI route passes	A PID, causal evidence, or immunity to MI-estimator failure

This table is scoped to the present Prisoma–pid-rs interaction and the reviewed high-value research candidates. It is not an exhaustive list of PID or adjacent information-decomposition research. Absence is not rejection. Any unlisted published or project-defined construction—including alternative redundancy or unique-information families, partial-entropy or integrated-information decompositions, and mixed-law, conditional, temporal, dynamic, manifold, neuromorphic, or differentiable programs—enters PID-M0 preserve as an `unassessed_literature_candidate`. Preserve its exact intake before deciding applicability. Do not relabel an adjacent decomposition as an ordinary target PID, and do not alias an unassessed construction to a listed functional because its atom names look familiar.

The functional ID does not identify one scalar output. Every PID result must also name:

- `quantity_id`, such as a cumulative shared-exclusion value or a Möbius-inverted atom;
- `lattice_coordinate`, including the exact antichain or node and ordered source roles;
- `component_kind = net | informative | misinformative` when the functional has that split; and
- `aggregation_scope = pointwise | averaged` with the averaging law when applicable.

A non-lattice information hierarchy must instead name its exact output family and index, such as per-source unique information or `SE_K`. It must not invent a lattice coordinate so that its result resembles a full PID. An objective result must name the exact composition family, coefficient vector, non-PID terms, optimization direction, and the identities of every input quantity.

A cumulative lattice value is not its Möbius-inverted atom. A net atom is not either of its nonnegative informative or misinformative components. A pointwise realization is not its probability-law average. Equal field names or familiar labels such as “redundancy” do not make these coordinates interchangeable.

`project_defined` is an origin tag, not a generic scientific-object kind. A new functional receives a functional ID; a new estimator receives a route ID; a new diagnostic or composition receives its own kind-specific ID. A catch-all “extension” ID is forbidden because it would erase the distinction this contract is intended to preserve.

2.1 Typed relations preserve both distinction and lineage

Distinct objects can still have precise relationships. The registry must use typed edges rather than prose similarity or author-name lineage:

Edge	Meaning	Required evidence
<code>defines_quantity</code>	A functional defines one named cumulative value, atom, or component coordinate	Defining equations and exact reference locator
<code>targets_functional</code>	A sample estimator targets one functional and named quantity	Estimator definition and admitted law or sampling domain
<code>evaluates_functional</code>	A declared-law evaluator computes one functional and named quantity	Evaluator contract and admitted law schema
<code>implements_route</code>	A software surface implements one estimator or evaluator route	Software revision and conformance evidence
<code>recovers_on_domain</code>	One construction provably agrees with another on a stated subdomain	Mapping theorem, assumptions, and exact domain
<code>motivated_by</code>	A later object adopts an idea without claiming equality	Exact reference and bounded relationship statement
<code>validated_by</code>	A theorem, oracle, counterexample, or fixture tests a named claim	Evidence identity and covered coordinates
<code>composes_quantities</code>	An objective or diagnostic combines named coordinates	Coefficients, other terms, and evaluation route

Use `alias_of` only for definitionally identical identities. One route cannot alias several functionals. Shared authors, notation, lineage, or limiting intuition do not create a `recovers_on_domain` edge.

The 2017 neural-goal paper supplies a coordinate language for constructing and comparing local information-theoretic goals. It does not define one universal PID functional. The parthood work supplies semantic structure. It does not supply a finite-sample estimator. Infomorphic learning composes PID coordinates into objectives. The objective is downstream of the PID definition.

The Ehrlich paper also draws a boundary inside the shared-exclusions lineage. Its practical continuous formula is not invariant under arbitrary invertible re-encodings of individual source variables without a prescribed preprocessing choice. The paper therefore does not identify that formula as a direct concretization of the earlier measure-theoretic construction. Prisoma records the two functionals as related and distinct. A shared name or motivation is not a recovery theorem.

The Matthias et al. arXiv-v1 result supplies a three-source counterexample. It rules out one PID family that would satisfy local positivity, target chain rule, and re-encoding invariance for all source counts. It does not show that no PID exists at every fixed source count. It does not validate MGW, make all negative atoms necessary, or make PID useless. Its counterexample is discrete; the paper explicitly leaves open whether a family restricted to continuous laws can avoid the inconsistency.

2.2 Gaussianity is a law restriction, not a PID identity

"Gaussian PID" is not one method. The relevant research lines answer different questions:

- the bivariate $\sim$ -PID is the BROJA-PID. The Gaussian paper then defines a distinct restricted functional, $\sim_G$, by allowing only jointly Gaussian couplings Q in the BROJA optimization. Its unique terms upper-bound BROJA/ $\sim$, while its redundant and synergistic terms lower-bound them in general. Equality when the input law P is jointly Gaussian is a conjecture, not a mapping theorem. The covariance-law evaluator, sample-covariance plug-in estimator, and finite-sample bias-corrected estimator are three further route layers targeting $\sim_G$, not unrestricted BROJA/ $\sim$;
- the bivariate deficiency δ -PID is another functional. On Gaussian inputs, δ_G restricts the degrading channel to the linear additive Gaussian class and therefore upper-bounds unrestricted deficiency. The further convex construction defines $\bar{\delta}_G$, an exact surrogate functional whose relationship to δ is by proved bounds and extremal equalities. It is not merely numerical error in an exact δ evaluator. Agreement among δ , δ_G , $\bar{\delta}_G$, $\sim_G$, MMI, or another PID on a fixture does not make any pair aliases;
- the same 2023 unique-information paper separately defines a parameterized δ^λ family and the I-PID. The I-PID is proved Blackwellian for jointly Gaussian laws, while its general Blackwellian property is stated as a conjecture. Neither object is the BROJA/ $\sim$ -PID merely because the paper analyzes their relationships; and
- the Lyu–Clark–Raviv construction gives a conditional-independence information hierarchy with covariance-law formulas and a plug-in estimator. For $N \geq 3$, the authors deliberately do not assign redundancy. Its unique-information and synergy-spectrum quantities therefore must not be padded with an invented redundancy, called a complete antichain decomposition, or pooled with atoms from another PID.

For every Gaussian route, bind the exact functional or hierarchy quantity, source count, covariance domain, positive-definiteness or regularization conditions, analytic-law versus sample route, and all estimator parameters. Ridge regularization and bias correction belong to route identity; neither silently changes the population functional. Empirical bell-shaped marginals do not prove the required joint Gaussian law. Atomic spike counts, quantized variables, singular manifolds, dependent episode rows, and arbitrary VLA embeddings do not become eligible merely because a covariance matrix can be computed. Such data require a matching law, estimator regime, and application gate or an explicit multi-estimand sensitivity study.

The Lyu hierarchy needs one further route distinction. Its order-specific synergistic effects can be negative, so their names do not make them nonnegative PID atoms. A fixed positive ridge makes a regularized finite-sample route and does not retain the paper's exact blockwise affine invariance. If a ridge sequence vanishes with sample size, the limit claim needs its own rate and conditioning assumptions. Record the ridge rule, not only the realized covariance matrix.

Bias correction is also route-specific evidence, not a theorem that the corrected atoms are unbiased. A later sampling study reports materially different bias across PID components and treats its own corrections as heuristic despite large-sample analysis. The 2023 $\sim_G$ correction therefore remains a named research route requiring independent matched-regime bias, variance, coverage, and failure studies. It is not the automatic preferred route and does not estimate unrestricted BROJA/ $\sim$ unless the unproved Gaussian-optimum equality happens to hold.

2.3 Bivariate unique-information identities

For target M and sources X, Y , define

$$\Delta_P = \{Q_{MXY} : Q_{MX} = P_{MX}, Q_{MY} = P_{MY}\}.$$

Canonical source expression: $\Delta_P = \{Q_{MXY} : Q_{MX} = P_{MX}, Q_{MY} = P_{MY}\}.$

The BROJA/~PID fixes one unique-information coordinate by

$$\widetilde{UI}(M : X \setminus Y) = \min_{Q \in \Delta_P} I_Q(M; X | Y).$$

Canonical source expression: $\widetilde{UI}(M : X \setminus Y) = \min_{Q \in \Delta_P} I_Q(M; X | Y).$

The three bivariate PID consistency equations then determine redundancy, the other unique term, and synergy. “Union information” is an equivalent coordinate of this same decomposition, not a second functional.

The Gaussian-restricted construction changes the feasible set:

$$\widetilde{UI}_G(M : X \setminus Y) = \min_{Q \in \Delta_P \cap \mathcal{G}} I_Q(M; X | Y),$$

Canonical source expression: $\widetilde{UI}_G(M : X \setminus Y) = \min_{Q \in \Delta_P \cap \mathcal{G}} I_Q(M; X | Y),$

where $\mathcal{G}$ is the admitted jointly Gaussian coupling class. Consequently $\widetilde{UI}_G \geq \widetilde{UI}$; the corresponding $\sim_G$ redundant and synergistic terms are lower bounds on the BROJA/~ terms. The paper conjectures equality when the input law P_{MXY} is jointly Gaussian. It does not prove it. Thus $\sim_G$ is a separately named functional related to BROJA/~ by bounds and a conjecture, not a law evaluator already known to return the BROJA/~ atoms.

The deficiency δ -PID instead measures the cost of approximating the $M \rightarrow X$ channel through Y :

$$\delta(M : X \setminus Y) = \inf_{P_{X'|Y}} \mathbb{E}_M[D_{KL}(P_{X|M} \| P_{X'|Y} \circ P_{Y|M})].$$

Canonical source expression: $\delta(M : X \setminus Y) = \inf_{P_{X'|Y}} \mathbb{E}_M[D_{KL}(P_{X|M} \| P_{X'|Y} \circ P_{Y|M})].$

Restricting $P_{X'|Y}$ to linear additive Gaussian channels defines δ_G , so $\delta_G \geq \delta$. The paper then defines the further convex surrogate δ^*_G , with $\delta^*_G \geq \delta_G \geq \delta$. After the paper’s redundancy symmetrization, its δ^*_G unique atoms upper-bound, and its redundancy and synergy atoms lower-bound, the unrestricted δ -PID atoms; proved extremal cases agree. “Approximate” describes this mathematical surrogate relationship. A numerical implementation can still have additional solver error and needs a separate evaluator identity.

The 2023 δ^λ family combines approximation divergence and conditional-information leakage in a parameterized objective,

$$\delta^\lambda(M : X \setminus Y) = \inf_{P_{X'|MY}} \{ \mathbb{E}_M[D_{KL}(P_{X|M} \| P_{X'|M})] + \lambda I(M; X' | Y) \},$$

Canonical source expression: $\delta^\lambda(M : X \setminus Y) = \inf_{P_{X'|MY}} \{ \mathbb{E}_M[D_{KL}(P_{X|M} \| P_{X'|M})] + \lambda I(M; X' | Y) \},$

with the induced-channel and symmetrization conventions fixed by the paper. The paper states that the $\lambda \rightarrow \infty$ and $\lambda \rightarrow 0$ regimes yield the deficiency and $\sim$ -PIDs, respectively. The displayed raw scalar requires more care: when an exact copy is feasible, choosing $X'=X$ gives

$$0 \leq \delta^\lambda(M : X \setminus Y) \leq \lambda I(M; X | Y).$$

Canonical source expression: $0 \leq \delta^\lambda(M : X \setminus Y) \leq \lambda I(M; X | Y).$

Thus $\delta^\lambda \rightarrow 0$ as $\lambda \downarrow 0$ whenever the conditional information is finite. The normalized value δ^λ/λ or a lexicographically selected optimizer can approach the $\sim$ -PID objective. Prisoma therefore records the small- λ relationship as a paper claim needing a normalization, existence, and limit theorem before adding `recovers_on_domain`. Each $\lambda > 0$ is part of functional-instance identity; $\lambda = 0$ is a degenerate endpoint, not an alias for BROJA/ $\sim$.

The separate I-PID starts from information deficiency,

$$\delta^I(M : X \setminus Y) = \sup_{P_{T|M}} (I(T; X) - I(T; Y)),$$

Canonical source expression: $\delta^I(M : X \setminus Y) = \sup_{P_{T|M}} (I(T; X) - I(T; Y)).$

and symmetrizes this quantity into a bivariate PID. Its optimizer T is intended to represent the part of M uniquely accessible from one source. The paper proves the resulting I-PID Blackwellian for jointly Gaussian P_{MXY} and labels the unrestricted Blackwellian statement a conjecture. That theorem is a property of I-PID; it does not turn I-PID into BROJA, δ , or a Gaussian sample estimator.

3. Result identity is mandatory

Every serialized or published PID result, named information-hierarchy quantity, or PID-derived objective result must bind this identity tuple:

```
scientific_object_kind
scientific_object_id
functional_instance = Unparameterized
                      | Parameterized { exact_parameters, instance_id }
defining_references_and_versions
quantity = Cumulative { quantity_id, lattice_coordinate_or_antichain }
           | MobiusInvertedAtom { quantity_id, lattice_coordinate_or_antichain }
           | NamedFunctionalQuantity { quantity_id, output_indexing_structure }
           | ObjectiveValue { composition_family_id, objective_instance_id }
component_kind = net | informative | misinformative | not_applicable
source_variables_in_order
target_variable
source_count_and_lattice_or_output_structure
input_law_kind
declared_law_identity_or_sample_and_split_identity
route = sample_estimator | declared_law_evaluator
route_id_and_revision
route_parameters_and_configuration
transform_identity_and_fit_scope
fit_evaluation_row_relation
population_and_support_contract
gauge_metric_and_scale_when_applicable
zero_mass_null_event_and_boundary_conventions
information_unit_and_log_base
aggregation = Pointwise { realization_identity }
              | Averaged { averaging_law }
sign_convention
sampling_unit_dependence_and_resampling_design
uncertainty_procedure_and_multiplicity_family
software_and_feature_identity
validation_scope_and_status
application_gate_status
objective_composition = NotApplicable
                      | Composition { family_id, coefficient_vector,
                                      input_quantity_identities, non_pid_terms,
                                      optimization_direction }
```


quantity and aggregation are tagged choices. Do not serialize construction or averaging again as free strings that can contradict the selected variant. A route may support several variants, but one result instance names exactly one variant and coordinate.

route is a tagged choice. A result cannot claim both `sample_estimator` and `declared_law_evaluator`. Evaluation kind must be derived from that variant, not repeated as a possibly contradictory string.

The same normalized categorical PMF can arise from different scientific inputs. Sample counts produce a plug-in estimate of an unknown population law. Exact rational masses can specify an analytic law. The functional values can agree while sample identity, occupancy, uncertainty, and claim scope remain different. Nominal types must preserve that distinction.

4. Mathematical non-transfer rules

4.1 PID equations do not choose a measure

For two sources,

$$I(S_1, S_2; T) = R + U_1 + U_2 + S,$$

Canonical source expression: $I(S_1, S_2; T) = R + U_1 + U_2 + S,$

$$I(S_1; T) = R + U_1, \quad I(S_2; T) = R + U_2.$$

Canonical source expression: $I(S_1; T) = R + U_1, \quad I(S_2; T) = R + U_2.$

These equations do not identify R. A redundancy principle chooses a PID measure. Two atom vectors that satisfy the equations can still answer different scientific questions.

4.2 Quantization changes the estimand

A fitted quantizer maps numeric variables to new categorical variables. MGW on those variables is not the Ehrlich continuous functional of the original numeric variables. Binning sensitivity is therefore an estimand sensitivity, not only an estimator robustness check. No categorical-to-continuous limit is claimed without a theorem.

4.3 Negative atoms and signed components are not semantic labels

MGW informative and misinformative components are formal surprisal components. “Misinformative” does not mean wrong, deceptive, unsafe, or harmful. Net atoms can be negative. Never clamp them unless the named functional itself defines the operation. Never translate a sign into an application meaning without independent theory and intervention evidence.

4.4 Transform invariance is functional- and route-specific

Population mutual information can be invariant under suitable invertible transformations. A finite-sample estimator need not be. The Ehrlich continuous construction also exposes source gauge and relative-scale choices. Scaling, normalization, PCA, PLS, pooling, quantization, and learned projection must appear in result identity. Stability under one transform is not a general invariance theorem.

4.5 Exact finite-law evidence has a bounded scope

An integer-count certifier can prove facts for its admitted positive count laws and arithmetic bounds. It does not certify arbitrary binary64 neural probabilities. A rational-law extension can certify its own admitted schema. A binary64 evaluator remains a deterministic numerical evaluator, not an exact certifier, even when it agrees on bounded fixtures.

A declared-law evaluator must reject non-unit mass. Convenience normalization is a separately named transform with its own receipt because it changes the declared input. A soft specified law must be evaluated directly; drawing target samples and evaluating the induced empirical law does not evaluate the original soft law.

The planned fixed-source-law MGW fixture tests a paper-derived identity under one frozen alphabet, event map, source order, antichain lattice, log base, positive source support, and exact source marginal. The MGW informative cumulative term depends only on the matching source union-event probability. Averaging therefore depends only on the exact source marginal. Möbius inversion on the same lattice preserves equality of the informative atom vector. Under those conditions, $\text{delta}(\text{net_atom}) = -\text{delta}(\text{misinformative_atom})$ for every coordinate. This statement does not survive a changed source marginal, event map, lattice, support convention, or component identity. Until an exact theorem locator is pinned, describe the fixture as project-defined validation of a paper-derived algebraic identity. Do not promote the fixture itself to a paper-defined theorem.

4.6 Higher source count is a different mathematical regime

The source count and antichain lattice are part of the fully instantiated mathematical result and admitted domain, even when one stable functional-family ID spans several source counts. A two- source oracle does not validate three- or four-source atoms. Higher-order categorical MGW work remains valuable, but it needs source-count-specific reconstruction, exact-law fixtures, complexity bounds, and interpretation. A pairwise screen is not a higher-order PID.

4.7 Cross-PID studies are multi-estimand studies

Comparing different PID constructions can be scientifically valuable. It must be designed as a multi-estimand study, not as a search for whichever measure gives the preferred answer. Freeze each functional, coordinate, route, assumptions, hypothesis, and interpretation before outcomes are inspected. Pass each route's own gates, report disagreement, and control any family of claims.

Never average atom vectors across functionals, vote them into a “consensus PID,” select a measure after seeing its sign, or treat numerical agreement as a mapping theorem. Agreement is one observed comparison result. Disagreement is not an implementation failure unless an applicable theorem or oracle requires equality.

5. Applicability decision procedure

Choose the route from the scientific question and population law. Do not choose it from which API returns a number.

Declared problem	Admissible route	Current disposition
Exact finite categorical law	Direct MGW declared-law evaluation; rational or interval assurance when admitted	Upstream request; not in the Prisoma pin
Sampled categorical rows	MGW empirical-PMF plug-in estimator with sampling and occupancy diagnostics	Available at the pin
Sparse positive empirical counts	Bounded MGW count-law plug-in route sharing the row kernel	Highest-value upstream request
Numeric data deliberately mapped to categories	Fitted categorical estimand with transform and row-relation receipts	Available only as a descriptive warning route at the pin
Purely continuous, full-dimensional, finite-information tuple	Ehrlich functional through its named kNN estimator and gauge contract	Default-off; application gate blocked
Exactly specified jointly Gaussian two-source vector law	One predeclared $\sim_G$, δ_G , or $\bar{\delta}_G$ functional plus its exact declared-law route; unrestricted BROJA/ $\sim$ or δ may be claimed only with an applicable equality theorem	Preserved research candidates; no current Prisoma route
Sampled rows under a predeclared jointly Gaussian two-source model	One named $\sim_G$ sample route, with plug-in versus bias-corrected identity fixed before outcomes; a sample-covariance route is not the population evaluator	Preserved research candidates; no current Prisoma route
Exactly specified or sampled jointly Gaussian N-source law	One named Lyu–Clark–Raviv hierarchy quantity through its law evaluator or plug-in estimator	Research candidate; for $N \geq 3$, not a complete PID
Deterministic continuous target path or singular joint law	No current continuous PID route	Abstain or define a different estimand
Mixed, atomic, quantized, singular, or unknown law	A separately defined matching functional and estimator	No automatic route; research definition required
Independent episodes or clusters	Group-preserving schedule appropriate to the statistic	Upstream schedule substrate requested
High-dimensional transformed VLA tensors	No active PID interpretation until matched-regime validation passes	Current NO-GO/blocked boundary

Declared problem	Admissible route	Current disposition
Local neural training goal	Named objective composition plus a separate evaluation/gradient route	Future work; never call the objective a PID measure

The decision procedure is:

- 1 Freeze the scientific question, target population, source order, target, and timing.
- 2 State the law class. Include deterministic ancestry, mixed support, and dependence.
- 3 Choose one functional or named information hierarchy whose domain and axioms match the question.
- 4 Choose one evaluator or estimator that targets that object and exact quantity.
- 5 Bind every applicable lattice coordinate or hierarchy index, component, aggregation law, transform, fit relation, gauge, unit, and source-count output structure.
- 6 Pass the applicable population, measure, route, and application gates.
- 7 Compare against simpler non-PID quantities and strong prediction baselines.
- 8 Publish the complete denominator, abstentions, warnings, and negative controls.

If step 2 is unresolved, stop. A method name cannot repair an unidentified population law.

6. Four gates, with route-specific meaning

Prisoma's H3 sample analysis retains four independent gates:

- 1 **Population gate.** The quantity is defined, finite, and relevant for the declared population.
- 2 **Measure gate.** The functional or hierarchy's mathematical commitments fit the scientific question.
- 3 **Estimator gate.** The sample estimator recovers that named object and quantity in the frozen regime with adequate bias, failure detection, and uncertainty behavior.
- 4 **Application gate.** The real variables and sampling process lie inside the validated regime.

A declared-law evaluator has no finite-sample estimator claim. Replace gate 3 with an **evaluator-correctness gate** that checks implementation, canonicalization, arithmetic, and an independent oracle. This substitution does not clear the population or application gates. An exact certifier supplies assurance only for the facts and inputs named by its certificate.

Passing one gate never passes another. KSG MI validation does not validate Ehrlich redundancy. Categorical exact assurance does not validate continuous shared exclusions. A low-dimensional fixture does not validate high-dimensional VLA embeddings. Stable output does not establish a correct estimand.

7. Dependence, row identity, and uncertainty

7.1 Row relations

Matrix hashes do not prove that transform-fit rows and evaluation rows are disjoint. Compute row relations from caller-declared stable sampling-unit identities. The initial contract must reject duplicate identities. It can then distinguish:

- the same ordered unique sequence;
- the same unique identity set in another order;
- partial overlap with an exact shared count; and
- disjoint unique identity sets.

If repeated identities are scientifically meaningful, add a later multiset or occurrence-aware contract. Do not call a repeated collection an unordered set. A caller assertion remains an assertion. It must never inhabit a computed-proof variant.

A row identity and a group identity are different fields. Every observed row receives one unique row identity; several rows may carry the same episode or cluster identity. If the independent sampling unit spans multiple rows, record that hierarchy and use it for splitting and uncertainty. Never substitute a repeated episode ID for the unique row ID merely to obtain a relation result.

7.2 Group schedules

A schedule API can prevent accidental splicing of independent episodes. It does not, by itself, make every statistic valid. Sampling whole groups with replacement duplicates all numeric rows in a selected episode. Distinct occurrence IDs preserve provenance but do not change the duplicated coordinates. The pinned continuous KSG/Ehrlich route rejects exact ties.

The first continuous-compatible episode route should therefore be a prespecified without- replacement group subsampling diagnostic, a per-group statistic, or a future weighted/cluster- aware estimator. It must state its target and must not claim bootstrap confidence-interval calibration. Replacement schedules may remain useful for categorical statistics and other callbacks that admit duplicated rows. A continuous callback must abstain when a realized schedule violates its sample contract.

7.3 Tail probabilities

A nominal Monte Carlo permutation p-value and a stationary-surrogate tail fraction are different types. A nominal p-value must bind its hypothesis and predeclared family; the null-invariance group or randomized-assignment mechanism that licenses the transformation; any orbit conditioning; the draw scheme, count, and random-stream identity; the tail rule; and the finite-sample correction. Exact enumeration, conditional randomization, and Monte Carlo approximation are different statuses. A circular-shift score cannot acquire p-value status from the same arithmetic.

Only typed nominal p-values may enter BH or BY adjustment, and admission by type does not establish the adjustment theorem's assumptions. BH requires a justified independence or applicable positive- dependence regime (or a cited extension that covers the declared family). BY can address arbitrary dependence among valid p-values under its own finite-family procedure; it does not repair an invalid null, an invalid transformation group, or a post-selected family.

8. Current pin versus upstream roadmap

The following table prevents future work from being written as if it already ships in Prisoma.

Capability	pid-rs@796c11e	Prisoma status
Row-based empirical categorical MGW	Present	Active descriptive route
Fitted equal-width categorical MGW	Present	Same-row warning route; no H3 evidence
Continuous Ehrlich kNN report route	Present behind experimental surface	Application-blocked
Williams–Beer I_min	Present	Preserved inactive comparator, never fallback
Separate functional/quantity registry and typed graph edges	Absent	Upstream Wave 0 request
Borrowed sparse empirical-count MGW2 API	Absent	Upstream priority
Specified rational-law evaluator/certificate schema	Absent	Later upstream priority
Declared binary64 finite-law evaluator	Absent	Only after count/rational agreement
Computed fit/evaluation row-relation receipt	Absent	Upstream priority
Public checked component sum with typed memory/work meaning	Absent	Upstream priority
Nominal p-value and surrogate-tail nominal types	Absent; calibration is recorded in one shared report/value shape	Upstream priority
Episode/group schedule contract	Absent	Experimental upstream request
Complete tuple-level Ehrlich support/gauge contract	Incomplete	Experimental upstream request
Neutral pid-runlog decision event	Absent in schema 2	Orthogonal upstream request

Capability	pid-rs@796c11e	Prisoma status
Bivariate BROJA/∼, restricted ∼_G, deficiency δ, restricted δ_G, and convex-surrogate δ_G functionals and routes	Absent	Preserved as distinct research identities; no runtime request yet
Bivariate δ^λ family and information-deficiency I-PID	Absent	Preserved research identities; no runtime request yet
Lyu–Clark–Raviv Gaussian hierarchy evaluator/plugin routes	Absent	Preserved research identities; no runtime request yet

The current world-model reference therefore continues to store forecast commitments and execution receipts in strictly named schema-2 `label_observed` compatibility envelopes. They are not outcome labels. Prisoma must migrate only after a new run-log schema is adopted and historical replay is preserved.

9. Upstream priority order

The best sequence is:

- 1 add functional and quantity-coordinate identities plus typed graph edges while preserving current method IDs as canonical routes;
- 2 add `ResourceEstimate` construction and checked component sums with typed memory/work meaning;
- 3 add unique-row identity relations and typed nominal p-value/surrogate outputs;
- 4 refactor one private canonical weighted positive-law kernel;
- 5 expose a borrowed, bounded sparse empirical-count MGW2 view and a distinct empirical result;
- 6 add the fixed-source-law MGW fixture and bridge admitted counts to the existing exact certifier;
- 7 add a distinct specified-rational-law schema and certificate path;
- 8 add a binary64 declared-law evaluator only after the count and rational routes agree;
- 9 add group schedule infrastructure with explicit callback admissibility;
- 10 strengthen the continuous Ehrlich tuple-law and gauge contract; and
- 11 represent infomorphic objectives as downstream typed compositions outside estimator identity.

Do not add BROJA or `I_min` to Prisoma's active path. Do not prioritize GPU or MPS work before identity, mathematics, assurance, and dependence are correct. Do not add mixed, conditional, or temporal functionals by extending an existing name. Each requires its own definition and program.

The first resource composition is a checked component-wise aggregate, not one homogeneous physical quantity. Its contract means every listed component executes and its memory may coexist. Summed `estimated_bytes` is then a conservative co-resident memory upper bound, not an exact sequential peak. Summed `pairwise_distances` and `operations_hint` are additive work charges, not simultaneous resources. Alternatives and mutually exclusive branches need a different plan type. A later phase plan may take memory maxima only after retained outputs and transient scratch have different types and explicit lifetimes.

10. Preserve and promote novel PID research

Novel valid work is an asset for the thesis and future papers. Record its origin separately from its implementation maturity, evidence level, and current disposition. These axes are orthogonal. For example, a paper-defined functional can be application-blocked, while a project-defined evaluator can have strong exact-oracle evidence.

Origin	Meaning	Allowed claim
<code>paper_defined</code>	Exact quantity or estimator defined by a cited paper	The cited scope only
<code>paper_derived</code>	New implementation or composition derived from cited mathematics	Repository derivation, not new theory by default
<code>project_defined</code>	New diagnostic, contract, evaluator, estimator, or objective proposed here	Explicit project origin and unvalidated status

Origin	Meaning	Allowed claim
external_reference_code	Separately maintained implementation used for a named comparison	That bounded comparison only
no_implementation	Literature object or request with no repository route	Definition and gap only

Record disposition on a second axis:

Disposition	Meaning
active_route	Eligible for the named analysis after every gate passes
preserved_comparator	Retained for a named comparison; never an automatic fallback
research_candidate	Incomplete mathematical, software, or empirical program
application_blocked	May support methods work but not the current application claim
negative_boundary	Reproducible failure of a stated theorem, oracle, route, or application gate
superseded_preserved	Replaced by a correction while the original record remains immutable

Every new PID-related object needs:

- a mathematical definition and law domain;
- exact output quantity, coordinate construction, component, and aggregation;
- source count, lattice, and source/target order;
- stated axioms, invariances, and known trade-offs;
- a proof or derivation ledger;
- a separate evaluator or estimator identity;
- an independent oracle or counterexample family;
- resource and determinism bounds;
- sampling, dependence, and uncertainty assumptions;
- typed failure and abstention behavior;
- comparison with the nearest existing functional without pooling atoms;
- an application hypothesis that simpler MI or prediction baselines cannot already answer; and
- a publication claim that does not exceed the evidence level.

Research routes worth preserving include higher-source categorical MGW, exact and interval finite- law assurance, specified rational and binary64 law evaluation, dependence-aware schedules, continuous tuple/gauge contracts, bivariate BROJA/ $\sim$, $\sim_G$, δ , δ_G , δ^G , δ^λ , and I-PID programs, the distinct conditional-independence Gaussian hierarchy, and typed infomorphic compositions. Mixed-law, conditional, temporal, manifold, neuromorphic, and new differentiable PID proposals may also be valuable. They stay distinct research programs until their definitions, mapping theorems, estimators, and oracles exist.

A failed application gate removes a route from an active application claim. It does not erase a valid functional, implementation, theorem, negative result, or research artifact. If a derivation is wrong, retain the superseded bytes and publish the correction. Do not overwrite the record.

11. PID publication evidence ladder

Use the strongest achieved level and name all missing higher levels. The PID-P* labels below are local to this PID evidence ladder. They are not the repository's priority labels and are not the ecosystem E0–E5 capability scale.

Level	Evidence	Does not establish
PID-P0 definition	Exact object, law, equations, and provenance	Correct implementation
PID-P1 derivation	Reviewed proof or derivation obligations	Numerical correctness

Level	Evidence	Does not establish
PID-P2 implementation	Deterministic bounded code and conformance tests	Functional recovery
PID-P3 oracle	Independent exact, analytic, or cross-implementation fixtures	Planned-regime estimator validity
PID-P4 route regime	Estimator bias/coverage at matched scale and dependence, or evaluator error/conditioning across its admitted law domain; both include failure detection	Real-application validity
PID-P5 application	Frozen application gate and held-out incremental value over baselines	Causal mechanism or transport
PID-P6 intervention /replication	Target-engaging intervention and independent family/site replication	Universal validity

Every PID paper packet must contain:

- 1 the complete result-identity tuple;
- 2 the defining equations and full-team primary references;
- 3 the population, measure, route, and application verdicts;
- 4 source/target ancestry and timing;
- 5 transform-fit and evaluation row receipts;
- 6 exact software, feature, data, split, and environment identity;
- 7 oracle, negative-control, and matched-regime results;
- 8 dependence-aware uncertainty and the complete multiplicity family;
- 9 all requested, produced, warned, and abstained denominators;
- 10 strong non-PID baselines at matched information access and compute;
- 11 retained null, negative, and contradictory results; and
- 12 a claim table that separates functional, coordinate, estimator, application, and causal conclusions.

Potential publication contributions must stay distinct:

- a methods/assurance paper can study the scientific-object graph, sparse laws, exact bridges, and misuse-resistant types;
- a categorical paper can study one exact fixed-law MGW property and its bounded evaluator;
- a continuous methods paper can study a matched Ehrlich estimator regime and abstention boundary;
- an application paper can test held-out incremental value only after the first three gates pass;
- a rigorous negative paper can localize why a popular application regime is undefined, non-identifiable, or estimator-invalid.

None of these requires pretending that all PIDs agree.

12. Ecosystem contract

Every consumer must negotiate exact capability, not a family nickname. The handshake must bind the functional or hierarchy, quantity and lattice coordinate or output index, component, aggregation, route and parameters, source count, input-law kind, units, sampling and uncertainty design, software revision, features, and gate status. An objective consumer additionally binds the complete coefficient vector and every input-quantity identity. A Galadriel result and a Prisoma result that use the same `pid-rs` route are correlated uses of one implementation family. They are not independent replications merely because two applications emitted them.

Cross-project comparison is admissible only when the result-identity tuple matches or a mapping theorem defines the conversion. Otherwise, report a multi-estimand comparison. Never pool the atoms. Never let a consumer relabel an abstention as zero or route it to a different functional.

13. Twenty-question review before any method enters a claim

- 1 What exact functional or hierarchy and output quantity, lattice coordinate, or hierarchy index are being evaluated?

- 2 What law class is its mathematical domain?
- 3 Which equations, axioms, trade-offs, and typed relations to nearby objects define it?
- 4 What source count, lattice, source order, and target are fixed?
- 5 What units, aggregation law, component, null-event convention, and sign convention are used; for an objective, what exact input identities, coefficient vector, residual terms, and optimization direction define the instance?
- 6 Is the route an evaluator or an estimator?
- 7 What transform created the variables, and where was it fit?
- 8 What row identities prove the fit/evaluation relation?
- 9 What marginal and joint support assertions are required?
- 10 What gauge, metric, scale, regularization, and route parameters affect identity?
- 11 What dependence, sampling unit, resampling design, and multiplicity family define uncertainty?
- 12 Can resampling create ties or leave the estimator's admitted domain?
- 13 What independent oracle, theorem, or counterexample applies?
- 14 At what dimensions, sample sizes, signals, and support has the route been validated?
- 15 Which numerical, arithmetic, resource, and determinism bounds apply?
- 16 Which simpler MI, invariant, prediction, or causal baseline answers the same question?
- 17 What would falsify the functional, estimator, and application claims separately?
- 18 What interpretation is forbidden without intervention evidence?
- 19 Which ecosystem outputs are genuinely independent, which share code, fixtures, prompts, reviewers, or assumptions, and which load-bearing derivations and results has the human researcher independently reproduced and accepted under the applicable disclosure policy?
- 20 What exact evidence level can the paper claim, and which higher levels remain open?

Any unanswered question blocks a confirmatory PID claim. It does not block preserving the work as a typed research candidate.

14. Reproducible method-selection and publication process

The process is evidence. A final number without its decision path is not a reproducible PID result. Use the stages below for every new method, route, comparison, or application. The $PID-M^*$ labels name workflow stages. They are not the $PID-P^*$ evidence levels in Section 11.

14.1 Stage contract

Stage	Required work	Required output	Stop rule
PID-M0 preserve	Snapshot the proposal, defining papers, code, fixtures, and known failures before reinterpretation	Immutable intake manifest and origin record	Stop if the defining object or version cannot be identified
PID-M1 type	Assign separate IDs to the functional, quantity, evaluator, estimator, transform, certifier, validation artifact, objective, and interpretation	Scientific-object graph with typed edges	Stop if one ID spans nonidentical objects
PID-M2 prove	Re-derive the equations, domains, bounds, invariances, source-count scope, and limiting claims	Mathematics ledger with theorem locators, derivations, and counterexamples	Stop if an asserted mapping lacks a theorem or proof
PID-M3 admit	Match the population law, source and target ancestry, support, gauge, dependence, sampling unit, and transform	Applicability decision with explicit rejected routes	Stop if the law or timing is unresolved
PID-M4 design	Choose one evaluator or estimator, oracle, resource bound, uncertainty method, multiplicity family, and non-PID baselines	Frozen route and validation plan	Stop if the route cannot target the named quantity
PID-M5 freeze	Freeze hypotheses, coordinates, parameters, splits, warnings, abstentions, and claim language before outcomes	Content-bound analysis packet and amendment policy	Stop if outcome access predates the freeze
PID-M6 execute	Run the declared commands and retain all requested, produced, warned, abstained, and failed cases	Command log, environment identity, outputs, and complete denominator	Stop on identity drift or unplanned substitution
PID-M7 challenge	Recompute critical quantities and review mathematics, numerics, statistics, software, and application claims	Review dispositions, negative controls, and unresolved objections	Stop if a required objection is unresolved
PID-M8 publish	Build the paired Markdown and PDF artifacts and bind them to the evidence bundle	Publication packet, hashes, visual review, and claim table	Stop if source and publication view diverge

Do not skip a stage because an API returns a finite value. A stopped route remains a preserved research object. Record the exact blocker and the evidence needed to resume it.

14.2 Mathematics and applicability ledger

The mathematics ledger must contain:

- 1 the defining equation for every reported quantity;
- 2 the admitted probability-law class and source count;
- 3 every symmetry, invariance, bound, and sign claim used in interpretation;
- 4 a locator for each paper theorem and a separate label for each project derivation;
- 5 the exact construction from cumulative values to atoms or hierarchy outputs;
- 6 a counterexample for every tempting but invalid cross-method substitution;
- 7 numerical conditioning and arithmetic assumptions;
- 8 the evaluator or estimator target and its failure set; and
- 9 a route-by-route decision that says `admitted`, `blocked`, `deferred`, or `not_applicable`.

For a new derivation, include enough algebra for another researcher to reproduce it without the implementation. For a paper claim that appears inconsistent with its displayed objective, retain the paper statement and the project derivation as separate records. Do not repair the citation by silently changing the functional.

14.3 Execution and review packet

Every execution packet must bind:

- the complete result-identity tuple from Section 3;
- source, data, split, environment, software, feature, and configuration hashes;
- exact commands, exit status, stdout and stderr custody, and resource observations;
- the complete requested denominator and every abstention or failure reason;

- independent oracle results and deliberately failing controls;
- uncertainty and multiplicity receipts;
- comparison baselines with matched information access and compute;
- reviewer role, review scope, objection, disposition, and timestamp;
- reviewer and tool ancestry, including shared prompts, fixtures, implementations, model families, and any AI-assisted review or derivation that must not be counted as independent replication;
- a candidate-ownership record for every load-bearing result: the human researcher must be able to derive it, explain its assumptions and failure modes, reproduce its decisive checks, and comply with the university's disclosure and authorship rules; and
- amendment history without overwriting the earlier packet.

The reviewer must distinguish recomputation from rerunning the same implementation. Shared code, fixtures, or assumptions are shared ancestry. They are not independent replication.

14.4 Paired Markdown and PDF contract

The Markdown file is the canonical scientific source. The PDF is a deterministic publication view. The PDF must never become a second authority with hand-edited claims.

For each released pair:

- 1 render the PDF from the exact Markdown bytes with a versioned repository script;
- 2 record the source, renderer, and PDF SHA-256 digests and byte sizes;
- 3 record the renderer, Python, and font identities plus the exact command;
- 4 extract PDF text and check all required headings, equations, warnings, and references;
- 5 render every page to an image;
- 6 inspect every page for clipping, overlap, missing glyphs, broken equations, split tables, unreadable type, unexpected blank pages, and stale headers;
- 7 record page count and visual-review disposition; and
- 8 rerender after any source or renderer change.

A matching hash proves byte identity only. It does not prove readable layout or correct mathematics. Visual review does not replace text extraction or scientific review. The build receipt must state both results.

14.5 Publication decision

A positive paper claim needs the achieved $PID-P^*$ level, every missing higher level, and the exact claim boundary. A negative paper must identify the earliest failed stage, reproduce the failure, and state which nearby objects remain valid. A comparison paper must treat each PID as a separate estimand and retain disagreement.

Publication does not require every research route to become active. It requires every route to be named honestly. Preserve deferred, blocked, superseded, and negative work with enough information to resume, audit, or cite it.

15. Primary sources

- Williams and Beer, nonnegative decomposition and $I_{\min}$
- Wibral et al., PID coordinates for neural goal functions
- Gutknecht, Wibral, and Makkeh, parthood and formal logic
- Ehrlich et al., continuous shared exclusions and estimation
- Venkatesh and Schamberg, bivariate deficiency PID for multivariate Gaussian laws
- Venkatesh et al., restricted-Gaussian $\sim_G$ PID, estimation, and bias correction
- Lyu, Clark, and Raviv, conditional-independence Gaussian information hierarchies
- Schneider et al., trivariate infomorphic objectives
- Bertschinger et al., unique information and the BROJA construction
- Makkeh, Gutknecht, and Wibral, categorical shared exclusions
- Schick-Poland et al., general measure-theoretic PID
- Kraskov, Stögbauer, and Grassberger, k-nearest-neighbor mutual-information estimation
- Venkatesh, Gurushankar, and Schamberg, δ^λ , I-PID, and interpretations of δ and BROJA/ $\sim$
- Koçillari et al., component-specific PID sampling bias and heuristic corrections
- Makkeh et al., bivariate infomorphic networks
- Matthias et al., PID inconsistency results